Master Derivations — Pipeline of Coherence

All Formulas, Derivations, and Theorems
from the Kernel Repository

Framed by the Critical Complex Equilibrium $\mu = e^{i3\pi/4}$

Kernel Repository
github.com/beanapologist/Kernel

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Contents

A Proven Identities **4**

1 Fundamental Constants **4**
1.1 Theorem 3 — Critical Constant $\eta = 1/\sqrt{2}$ 4
1.2 Proposition 4 — Silver Conservation 5

2 The Critical Eigenvalue and Its Orbit **5**
2.1 Section 2 — Eigenvalue $\mu = e^{i3\pi/4}$ 5
2.2 Section 3 — Rotation Matrix $R(3\pi/4)$ 5

3 Quantum State and Balance **6**
3.1 Theorem 8 — Canonical Coherent State 6
3.2 Theorem 9 — Balance $\leftrightarrow$ Maximum Coherence 6

4 Coherence, Residual, and Lyapunov Functions **7**
4.1 Theorem 11 — Coherence Function 7
4.2 Theorem 12 — Palindrome Residual 7
4.3 Theorem 14 — Lyapunov–Coherence (Ohm–Coherence) Duality 8
4.4 Corollary 13 — Simultaneous Break 8

5 Trichotomy and the 8-Cycle **9**
5.1 Theorem 10 — Trichotomy 9
5.2 State Update Rule 9

6 Palindrome Precession and Torus Geometry **9**
6.1 The Palindrome Quotient . . . 9
6.2 Angular Increment and Torus Structure . . . 10
6.3 Theorem 5.1 — Zero Excess Resistance . . . 10
6.4 Theorem 6.1 — Optimality of $k = 1$ . . . 10

7 The $\Theta(\sqrt{n})$ Coherent Phase Detection **11**
7.1 NullSliceBridge: 8-Channel Phase Coverage . . . 11
7.2 Scaled Precession Phasor . . . 11
7.3 Dirichlet Kernel Analysis . . . 11
7.4 Effect of Decoherence on Scaling . . . 12
7.5 Constant-Factor Analysis . . . 12

8 Ohm–Coherence Duality: Extended Framework **12**
8.1 Single Channel . . . 12
8.2 Parallel Channels (MultiChannelSystem) . . . 13
8.3 Series Pipeline (PipelineSystem) . . . 13
8.4 Four-Channel Error Tolerance . . . 13
8.5 Jensen's Inequality Under Noise . . . 13
8.6 Qutrit Degradation . . . 14

9 Coherence Invariants and Interrupt Recovery **14**
9.1 Three Simultaneous Invariants . . . 14
9.2 Decoherence Severity Levels . . . 14
9.3 Coherence Recovery Formula . . . 14
9.4 Auto-Renormalization via Palindrome Residual . . . 15

10 Rotational Memory: $\mathbb{Z}/8\mathbb{Z}$ Addressing **15**

11 Noise Phase Transition and Universal Scaling Limit **15**
11.1 H_1 Coherence Phase Transition . . . 15
11.2 Universal Scaling Ceiling . . . 16
11.3 Stochastic Resonance: Optimal Coherence . . . 16

12 Arithmetic Zero-Overhead Periodicity **16**

B Empirical Observations **16**

13 Empirical Scaling and Complexity Verification **16**
13.1 Complexity Certificate and Regression . . . 17
13.2 Main Complexity Theorem (Continuous-Oracle Model) . . . 17

13.3 Empirical $\Theta(\sqrt{n})$ Scaling Plot . . . 18
13.4 Bank-Distribution Uniformity . . . 18
13.5 Stochastic Resonance: Optimal Coherence . . . 18
13.6 Empirical Verification of $\alpha_{\max} = 1 + 1/e$ . . . 19
13.7 Phase Asymmetry and Broken Rotational Symmetry . . . 20

14 KernelSync: Coherent Network Time Synchronisation 21
14.1 Phase Schedule and Setup . . . 21
14.2 Key Experimental Result . . . 21
14.3 Grover Analogy: EMA as Damped Diffusion . . . 22

15 Passive GHz Demo: Palindromic Precession Drive 22
15.1 Experimental Goal . . . 22
15.2 Phase Schedule . . . 22
15.3 Measurement Protocol . . . 23

C Conjectures and Hypotheses 23

16 Conjectures and Open Questions 23
16.1 Universal Scaling Ceiling ($\alpha_{\max}$) . . . 24
16.2 Residual-Based Heuristics . . . 24

17 Summary Table of All Theorems 25

18 Connection Between All Results 25

Critical Complex Equilibrium

$$\boxed{\mu \;=\; e^{i3\pi/4} \;=\; \frac{-1+i}{\sqrt{2}} \;=\; -\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}, \qquad |\mu| = 1, \quad \arg(\mu) = \tfrac{3\pi}{4}.} \tag{1}$$

μ is the unique eighth root of unity lying in the **second quadrant** of $\mathbb{C}$: negative real part $-1/\sqrt{2}$ and positive imaginary part $+1/\sqrt{2}$. It is the balanced eigenvalue at the heart of all constructions in this document. The entire theory emanates from μ and returns to it:

$\mu^8 = 1, \qquad \gcd(3,8) = 1 \;\Rightarrow\; \{1, \mu, \mu^2, \ldots, \mu^7\}$ partitions the unit circle into 8 equal slices of 45°.

This document collects, in logical order, *every* formula, derivation, and theorem from the Kernel repository. The exposition begins with the fundamental constants and eigenvalue structure, builds the coherence framework, derives the palindrome precession geometry, proves the $\Theta(\sqrt{n})$ complexity bound, presents the Ohm–Coherence duality, and concludes by returning to the critical equilibrium μ.

Part A

Proven Identities

1 Fundamental Constants

1.1 Theorem 3 — Critical Constant $\eta = 1/\sqrt{2}$

Theorem 1.1 (Critical constant)**.** *The equation $2\lambda^2 = 1$ has a unique positive root*

$$\eta \;=\; \frac{1}{\sqrt{2}} \;=\; \frac{\sqrt{2}}{2} \;\approx\; 0.7071067811865475. \tag{2}$$

This value satisfies $\eta^2 + \eta^2 = 1$ exactly, making it the unique amplitude for a balanced two-component state $(|\alpha|, |\beta|) = (\eta, \eta)$ with unit norm.

Proof. Substituting $\lambda = 1/\sqrt{2}$ gives $2(1/\sqrt{2})^2 = 2 \cdot \frac{1}{2} = 1$. Uniqueness follows since $f(\lambda) = 2\lambda^2$ is strictly increasing for $\lambda > 0$. □ □

1.2 Proposition 4 — Silver Conservation

Proposition 1.2 (Silver conservation)**.** *Define the* silver ratio $\delta_S = 1 + \sqrt{2}$ *and its conjugate* $\delta_S^{-1} = \sqrt{2} - 1$*. Then:*

$$\delta_S \cdot (\sqrt{2} - 1) = 1, \tag{3}$$

$$\delta_S^2 = 2\delta_S + 1, \tag{4}$$

$$1/\delta_S = \sqrt{2} - 1. \tag{5}$$

Proof. (3): $(1+\sqrt{2})(\sqrt{2}-1) = \sqrt{2}-1+2-\sqrt{2} = 1$. (4): $(1+\sqrt{2})^2 = 1+2\sqrt{2}+2 = 3+2\sqrt{2}$; $2(1+\sqrt{2})+1 = 3+2\sqrt{2}$. (5): follows from (3). □ □

Numerical values:

$$\delta_S \approx 2.41421356237, \qquad \delta_S^{-1} \approx 0.41421356237, \qquad \delta_S \cdot \delta_S^{-1} = 1.$$

2 The Critical Eigenvalue and Its Orbit

2.1 Section 2 — Eigenvalue $\mu = e^{i3\pi/4}$

Definition 2.1 (Balanced eigenvalue)**.** The *balanced eigenvalue* is

$$\mu \;=\; e^{i3\pi/4} \;=\; \cos\tfrac{3\pi}{4} + i\sin\tfrac{3\pi}{4} \;=\; -\tfrac{1}{\sqrt{2}} + \tfrac{i}{\sqrt{2}} \;=\; \eta(-1+i), \tag{6}$$

where $\eta = 1/\sqrt{2}$ (Theorem 1.1). It lies on the unit circle in the second quadrant.

Proposition 2.2. *The eigenvalue* μ *satisfies:*

$$|\mu| = 1, \tag{7}$$

$$\arg(\mu) = \tfrac{3\pi}{4}, \tag{8}$$

$$\mu^8 = 1 \quad (\textit{primitive 8th root of unity}), \tag{9}$$

$$\gcd(3,8) = 1 \;\Rightarrow\; \{1, \mu, \mu^2, \ldots, \mu^7\} \textit{ are all distinct.} \tag{10}$$

Proof. (7): $|\mu|^2 = \eta^2 + \eta^2 = 1$. (8): by definition. (9): $(e^{i3\pi/4})^8 = e^{i6\pi} = 1$. (10): since $\gcd(3,8) = 1$, multiplying by μ permutes all 8 residues of $\mathbb{Z}/8\mathbb{Z}$. □ □

2.2 Section 3 — Rotation Matrix $R(3\pi/4)$

Multiplication by μ on $\mathbb{C}$ corresponds to the real rotation matrix

$$R\left(\tfrac{3\pi}{4}\right) = \begin{pmatrix} \cos\frac{3\pi}{4} & -\sin\frac{3\pi}{4} \\ \sin\frac{3\pi}{4} & \cos\frac{3\pi}{4} \end{pmatrix} = \begin{pmatrix} -\eta & -\eta \\ \eta & -\eta \end{pmatrix}, \tag{11}$$

acting on $\mathbf{v} = (x, y)^\top \in \mathbb{R}^2$.

Proposition 2.3 (Rotation matrix properties)**.**

$$\det R = (-\eta)(-\eta) - (-\eta)(\eta) = \eta^2 + \eta^2 = 1, \tag{12}$$

$$R^\top R = I_2 \quad (\textit{orthogonality}), \tag{13}$$

$$R^8 = I_2 \quad (\textit{8-step periodicity}). \tag{14}$$

Proof. (12): direct computation. (13): R is an orthogonal matrix ($\det R = 1$, real, and $R^\top = R^{-1}$). (14): follows from $\mu^8 = 1$. □ □

The eight orbit points of μ at angles $3\pi k/4$, $k = 0, \ldots, 7$, are:

$$\{1,\, \mu,\, i,\, -\bar{\mu},\, -1,\, -\mu,\, -i,\, \bar{\mu}\} = \left\{e^{i3\pi k/4} : k = 0, \ldots, 7\right\}, \tag{15}$$

uniformly spaced at 45° intervals on the unit circle.

3 Quantum State and Balance

3.1 Theorem 8 — Canonical Coherent State

Theorem 3.1 (Canonical coherent state)**.** *The* canonical coherent state *is*

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad \alpha = \frac{1}{\sqrt{2}}, \quad \beta = \frac{e^{i3\pi/4}}{\sqrt{2}} = \frac{-1+i}{2}, \tag{16}$$

with $|\alpha|^2 + |\beta|^2 = 1$*. In component form,* $|\alpha| = |\beta| = \eta = 1/\sqrt{2}$*.*

Proof. $|\alpha|^2 = \eta^2 = 1/2$. $|\beta|^2 = |(-1+i)/2|^2 = (1+1)/4 = 1/2$. Sum equals 1. □ □

Definition 3.2 (Radius parameter)**.** For any normalised state (α, β) with $|\alpha| > 0$, the *radius parameter* is

$$r = \frac{|\beta|}{|\alpha|}. \tag{17}$$

At the canonical state, $r = 1$.

3.2 Theorem 9 — Balance ↔ Maximum Coherence

Theorem 3.3 (Balance-coherence equivalence)**.** *For a normalised state* (α, β) *with* $r = |\beta|/|\alpha|$*:*

$$|\alpha| = |\beta| = \frac{1}{\sqrt{2}} \iff C_{\ell^1} = 2|\alpha||\beta| = 1. \tag{18}$$

Equivalently, $r = 1$ *is the* unique *maximiser of* C_{ℓ^1}*.*

Proof. $C_{\ell^1} = 2|\alpha||\beta|$. By AM-GM, $2|\alpha||\beta| \le |\alpha|^2 + |\beta|^2 = 1$, with equality iff $|\alpha| = |\beta| = 1/\sqrt{2}$, i.e. $r = 1$. □ □

4 Coherence, Residual, and Lyapunov Functions

4.1 Theorem 11 — Coherence Function

Theorem 4.1 (Coherence function)**.** *The ℓ^1 coherence as a function of $r > 0$ is*

$$C(r) \;=\; \frac{2r}{1+r^2}. \tag{19}$$

Properties:

1. $C(1) = 1$ *(global maximum).*
2. $C(r) = C(1/r)$ *for all $r > 0$ (symmetric about $r = 1$).*
3. $\left.\frac{dC}{dr}\right|_{r=1} = 0, \quad \left.\frac{d^2C}{dr^2}\right|_{r=1} = -1 < 0$ *(strict local maximum).*
4. $0 < C(r) < 1$ *for all $r \neq 1$.*

Proof. Writing $C(r) = 2r/(1+r^2)$:

- $C(1) = 2/2 = 1$.
- $C(1/r) = (2/r)/(1+1/r^2) = (2/r)\cdot r^2/(r^2+1) = 2r/(1+r^2) = C(r)$.
- $C'(r) = [2(1+r^2) - 2r\cdot 2r]/(1+r^2)^2 = 2(1-r^2)/(1+r^2)^2$; at $r = 1$: $C'(1) = 0$.
- $C''(r) = -4r(3-r^2)/(1+r^2)^3$; at $r = 1$: $C''(1) = -4/4 = -1 < 0$. □

□

4.2 Theorem 12 — Palindrome Residual

Theorem 4.2 (Palindrome residual)**.** *Define*

$$R(r) \;=\; \frac{1}{\delta_S}\left(r - \frac{1}{r}\right), \qquad r > 0, \tag{20}$$

where $\delta_S = 1 + \sqrt{2}$ (Proposition 1.2). Then:

1. $R(r) = 0 \;\Leftrightarrow\; r = 1$.
2. $R(r) > 0$ *for all $r > 1$.*
3. $R(r) < 0$ *for all $0 < r < 1$.*

4. R is strictly monotone increasing on $(0, \infty)$.

Proof. $R(r) = 0$ iff $r = 1/r$ iff $r = 1$ (since $r > 0$). $R'(r) = (1/\delta_S)(1 + 1/r^2) > 0$ for all $r > 0$ (strictly increasing). □ □

4.3 Theorem 14 — Lyapunov–Coherence (Ohm–Coherence) Duality

Theorem 4.3 (Lyapunov–Coherence Duality)**.** *Let $\lambda = \ln r$ (the* Lyapunov exponent *of the state). Then:*

$$C(r) \;=\; \operatorname{sech}(\lambda) \;=\; \frac{1}{\cosh(\lambda)}, \tag{21}$$

with the identification

$$G_{\text{eff}}(\lambda) = \operatorname{sech}(\lambda), \qquad R_{\text{eff}}(\lambda) = \cosh(\lambda), \qquad G_{\text{eff}} \cdot R_{\text{eff}} = 1. \tag{22}$$

At the coherent fixed point $r = 1$ ($\lambda = 0$): $C = G_{\text{eff}} = 1$, $R_{\text{eff}} = 1$.

Proof. Using $\lambda = \ln r$, so $r = e^{\lambda}$:

$$C(r) = \frac{2r}{1 + r^2} = \frac{2e^{\lambda}}{1 + e^{2\lambda}} = \frac{2}{e^{-\lambda} + e^{\lambda}} = \frac{1}{\cosh \lambda} = \operatorname{sech}(\lambda). \quad \square$$

□

Corollary 4.4 (Symmetry via sech)**.** *$C(r) = C(1/r)$ follows immediately from* sech *being even:*

$$C(1/r) = \operatorname{sech}(\ln(1/r)) = \operatorname{sech}(-\ln r) = \operatorname{sech}(\ln r) = C(r).$$

Remark 4.5 (Inverse formula)*.* Given a coherence value $C \in (0, 1]$, the Lyapunov exponent is recovered via

$$\lambda = \operatorname{arccosh}\left(\frac{1}{C}\right) = \ln\left(\frac{1}{C} + \sqrt{\frac{1}{C^2} - 1}\right). \tag{23}$$

4.4 Corollary 13 — Simultaneous Break

Corollary 4.6 (Simultaneous break)**.** *The following three conditions are equivalent and hold simultaneously if and only if $r = 1$:*

$$r = 1 \iff C(r) = 1 \iff R(r) = 0 \iff \lambda = 0 \iff R_{\text{eff}} = 1. \tag{24}$$

Breaking any one condition breaks all others.

Proof. Follows directly from Theorems 4.1, 4.2, and 4.3: each of $C(r) = 1$, $R(r) = 0$, and $\lambda = 0$ uniquely characterises $r = 1$. □ □

5 Trichotomy and the 8-Cycle

5.1 Theorem 10 — Trichotomy

Define the *state orbit* under repeated multiplication by $\xi = r\mu$ (a scaled version of the balance primitive):

$$\xi^n = r^n \mu^n, \qquad n \in \mathbb{Z}_{\geq 0}. \tag{25}$$

Theorem 5.1 (Trichotomy)**.** *Let $\xi = r\mu$ with $r > 0$ and $|\mu| = 1$. Then:*

$$r = 1 \;\Rightarrow\; |\xi^n| = 1 \textit{ for all } n, \textit{ and the orbit } \{\xi^n\}_{n=0}^{7} \textit{ is a closed 8-cycle on the unit circle.} \tag{26}$$

$$r > 1 \;\Rightarrow\; |\xi^n| = r^n \to +\infty \textit{ (spiral out).} \tag{27}$$

$$r < 1 \;\Rightarrow\; |\xi^n| = r^n \to 0 \textit{ (spiral in / collapse).} \tag{28}$$

These three cases are mutually exclusive and exhaustive.

Proof. $|\xi^n| = r^n|\mu^n| = r^n$. For $r = 1$: $r^n = 1$ for all n; and $\mu^8 = 1$ closes the cycle. For $r > 1$ or $r < 1$: geometric sequence. □ □

5.2 State Update Rule

At each step, the state is advanced by

$$\beta \;\mapsto\; \mu\beta, \tag{29}$$

which preserves $|\beta|$ (hence r) while rotating the phase by $3\pi/4$. After 8 steps, β returns to its original value (mod global phase).

6 Palindrome Precession and Torus Geometry

6.1 The Palindrome Quotient

The integers 987 654 321 (descending digits) and 123 456 789 (ascending digits) are *digit-palindromes*. Their ratio gives

$$\frac{987\,654\,321}{123\,456\,789} = 8 + \frac{9}{123\,456\,789} = 8 + \frac{1}{13\,717\,421}, \tag{30}$$

since $9 \times 13\,717\,421 = 123\,456\,789$.

Proposition 6.1 (Two-palindrome complementarity)**.**

$$987\,654\,321 = 8 \times 123\,456\,789 + 9. \tag{31}$$

The integer part 8 *equals the* μ*-rotation period* ($\mu^8 = 1$, *Theorem 5.1); the fractional denominator* $D = 13\,717\,421$ *is the* slow-precession period.

6.2 Angular Increment and Torus Structure

Definition 6.2 (Precession increment)**.** The *palindrome precession increment* is

$$\Delta\Phi_0 = \frac{2\pi}{D} = \frac{2\pi}{13\,717\,421} \approx 4.578 \times 10^{-7} \text{ rad/step}. \tag{32}$$

The precession phasor at step n is

$$P(n) = e^{in\Delta\Phi_0} = \cos(n\Delta\Phi_0) + i\sin(n\Delta\Phi_0), \qquad |P(n)| = 1. \tag{33}$$

The state evolves under *two simultaneous periodicities*:

- **Fast 8-cycle:** $\beta \mapsto \mu\beta$, period 8 steps, angular step $3\pi/4$ per step.
- **Slow precession:** $\beta \mapsto P(n)\beta$, period $D = 13\,717\,421$ steps, angular step $\Delta\Phi_0$ per step.

These define coordinates on the torus $T^2 = S^1 \times S^1$ with winding numbers $(1/8,\ 1/D)$.

Theorem 6.3 (Torus super-period)**.** *The joint orbit closes after*

$$N_{\text{close}} = \text{lcm}(8,\ D) \times (\textit{winding alignment}) = 109\,739\,368 \textit{ steps}, \tag{34}$$

completing 41 152 271 *windings of the fast* S^1 *component.*

6.3 Theorem 5.1 — Zero Excess Resistance

Theorem 6.4 (Zero Excess Resistance)**.** *The palindrome precession step* $\beta \mapsto P(n)\beta$ *satisfies* $|P(n)| = 1$, *hence:*

$$|\beta| \textit{ is invariant} \;\Rightarrow\; r = 1, \quad C = 1, \quad R(r) = 0, \quad \lambda = 0 \textit{ are all preserved.} \tag{35}$$

There is zero excess resistance *(*$T = 0$*) compared to a zero-kick baseline.*

Proof. $|P(n)| = 1$ by construction (unit-norm phasor). Therefore $|\beta \cdot P(n)| = |\beta|$, so r is unchanged, and all four conditions of Corollary 4.6 are maintained. □ □

6.4 Theorem 6.1 — Optimality of $k = 1$

Theorem 6.5 (Optimality of $k = 1$)**.** *Among scaled precession rates* $\Delta\Phi(k) = 2\pi/(D \cdot k)$ *for* $k \geq 1$, *the choice* $k = 1$ *maximises per-step angular diversity at practical run lengths*

($\leq 2 \times 10^6$ *steps*). *For* $k > 1$, *the per-step phase shift* $\Delta\Phi(k) < \Delta\Phi_0$ *and diversity converges toward the zero-kick baseline as* $k \to \infty$.

7 The $\Theta(\sqrt{n})$ Coherent Phase Detection

7.1 NullSliceBridge: 8-Channel Phase Coverage

Definition 7.1 (NullSliceBridge)**.** The *NullSliceBridge* generates the eight unit-norm phasors

$$B_j = \mu^j, \qquad j = 0, 1, \ldots, 7. \tag{36}$$

Since $\gcd(3, 8) = 1$, the angles $\{j \cdot 135^\circ \bmod 360^\circ\}$ partition the unit circle into 8 equal slices of 45°. For any target angle θ, the nearest bridge angle is at most 22.5° away, guaranteeing overlap

$$\cos(22.5^\circ) \approx 0.9239 \tag{37}$$

between the best channel and the target.

7.2 Scaled Precession Phasor

To achieve a phase step of $\Delta\Phi = 2\pi/\sqrt{n}$ per coherent step, the slow phasor is scaled as

$$P_n(k) = P(k \cdot s_n), \qquad s_n = \left\lfloor \frac{D}{\sqrt{n}} \right\rfloor, \tag{38}$$

so that $P_n(k) = e^{ik\, s_n \Delta\Phi_0} \approx e^{ik \cdot 2\pi/\sqrt{n}}$. The composite probe phasor for channel j at step k is

$$\Pi_{k,j} = P_n(k) \cdot B_j. \tag{39}$$

7.3 Dirichlet Kernel Analysis

The amplitude accumulator for channel j after K steps is

$$A_j(K) = \sum_{k=0}^{K-1} G_{\text{eff}}(k)\, \cos\Big(\phi_{k,j} - \theta_{\text{target}}\Big), \tag{40}$$

where $\phi_{k,j} = k\Delta\Phi + j \cdot 3\pi/4$ and $G_{\text{eff}}(k) = \operatorname{sech}(\lambda_k)$.

When the KernelState remains coherent ($r = 1$, $G_{\text{eff}} = 1$), the sum for the best channel j^* reduces to the **Dirichlet kernel**:

$$A_{j^*}(K) = \frac{\sin(K\Delta\Phi/2)}{\sin(\Delta\Phi/2)} \cos\Big(\phi_{\text{mid}} - \theta_{\text{target}}\Big), \tag{41}$$

where $\phi_{\text{mid}} = \phi_{0,j^*} + (K-1)\Delta\Phi/2$. Denote

$$D_K(\Delta\Phi) = \frac{\sin(K\Delta\Phi/2)}{\sin(\Delta\Phi/2)}. \tag{42}$$

Lemma 7.2 (Linear growth of the Dirichlet kernel)**.** *For $\Delta\Phi = 2\pi/\sqrt{n}$ and $K \ll \sqrt{n}$:*

$$D_K\left(\frac{2\pi}{\sqrt{n}}\right) \approx \frac{K\sqrt{n}}{\pi}. \tag{43}$$

Proof. For small x, $\sin x \approx x$, so $\sin(\Delta\Phi/2) \approx \pi/\sqrt{n}$ and $\sin(K\Delta\Phi/2) \approx K\pi/\sqrt{n}$. □ □

7.4 Effect of Decoherence on Scaling

If the KernelState drifts with per-step radial perturbation ε, the Lyapunov exponent grows as $\lambda_k \approx k\varepsilon$, giving $G_{\text{eff}}(k) \approx e^{-k\varepsilon}$. The decoherent accumulator sum is

$$A_{j^*}(K) \approx \operatorname{Re}\left[e^{-i\delta}\frac{1-e^{-K(\varepsilon-i\Delta\Phi)}}{1-e^{-(\varepsilon-i\Delta\Phi)}}\right]. \tag{44}$$

For $\varepsilon \gg \Delta\Phi/K$, the sum saturates at $O(1/\varepsilon)$, and the detection threshold $0.15\sqrt{n}$ requires $\Omega(n)$ steps — *the $\sqrt{n}$ speedup is destroyed.*

7.5 Constant-Factor Analysis

$$c_{\text{theory}} = \frac{0.15\pi}{\langle\cos\delta\rangle} \approx 0.19, \tag{45}$$

where $\langle\cos\delta\rangle \approx 0.97$ is the expected cosine of the best bridge channel's angular error $\delta \in [0,\, 22.5°]$.

Empirical verification of c_{theory} against measured scaling data, including the full Complexity Certificate and regression details, is presented in Part B (Section 13).

8 Ohm–Coherence Duality: Extended Framework

Theorem 4.3 establishes $C = G_{\text{eff}} = \operatorname{sech}(\lambda)$ and $R_{\text{eff}} = \cosh(\lambda)$. This section develops the full circuit-analogy framework.

8.1 Single Channel

A *coherent channel* characterised by Lyapunov exponent $\lambda \geq 0$:

$$G_{\text{eff}}(\lambda) = \operatorname{sech}(\lambda) = \frac{1}{\cosh\lambda}, \qquad R_{\text{eff}}(\lambda) = \cosh(\lambda), \qquad G_{\text{eff}} \cdot R_{\text{eff}} = 1. \tag{46}$$

8.2 Parallel Channels (MultiChannelSystem)

For N parallel channels with Lyapunov exponents $\lambda_1, \ldots, \lambda_N$, conductances add:

$$G_{\text{tot}} = \sum_{i=1}^{N} G_{\text{eff}}(\lambda_i) = \sum_{i=1}^{N} \operatorname{sech}(\lambda_i), \qquad R_{\text{tot}} = \frac{1}{G_{\text{tot}}}. \tag{47}$$

For N identical channels ($\lambda_i = \lambda$):

$$G_{\text{tot}} = N \operatorname{sech}(\lambda), \qquad R_{\text{tot}} = \frac{\cosh \lambda}{N}. \tag{48}$$

8.3 Series Pipeline (PipelineSystem)

For M pipeline stages in series, resistances add:

$$R_{\text{tot}} = \sum_{j=1}^{M} R_{\text{eff}}(\lambda_j) = \sum_{j=1}^{M} \cosh(\lambda_j), \qquad G_{\text{tot}} = \frac{1}{R_{\text{tot}}}. \tag{49}$$

The bottleneck stage is $j^* = \operatorname{argmax}_j \cosh(\lambda_j)$.

8.4 Four-Channel Error Tolerance

The *four-eigenvalue structure* (4-channel redundancy) validates error tolerance:

$$\text{system coherent} \quad \Leftrightarrow \quad \#\Big\{ i : G_{\text{eff}}(\lambda_i) \geq \tau_{\text{thresh}} \Big\} \;\geq\; 3, \tag{50}$$

i.e., at least 3 of 4 channels must have $G_{\text{eff}} \geq 0.5$. This provides tolerance to one faulty channel.

8.5 Jensen's Inequality Under Noise

Theorem 8.1 (Jensen bound for OU noise)**.** *Let λ follow an Ornstein–Uhlenbeck process:*

$$d\lambda = -\theta(\lambda - \mu_{\text{OU}})\, dt + \sigma\, dW, \tag{51}$$

with parameters $\theta > 0$ (mean-reversion), μ_{OU} (mean), $\sigma \geq 0$ (noise amplitude). Since sech *is locally concave near $\lambda = 0$:*

$$\langle G \rangle = \langle \operatorname{sech}(\lambda) \rangle \;\leq\; \operatorname{sech}(\langle \lambda \rangle). \tag{52}$$

Noise degrades *average coherence below the coherence at the mean Lyapunov exponent.*

8.6 Qutrit Degradation

For a 3-level (qutrit) system, each pair of levels $\{0,1\}$, $\{0,2\}$, $\{1,2\}$ has its own Lyapunov exponent:

$$C_{\text{avg}} = \frac{\text{sech}(\lambda_{01}) + \text{sech}(\lambda_{02}) + \text{sech}(\lambda_{12})}{3}, \qquad C_{\min} = \min_{p \in \{01,02,12\}} \text{sech}(\lambda_p). \tag{53}$$

9 Coherence Invariants and Interrupt Recovery

9.1 Three Simultaneous Invariants

At the coherent fixed point $r = 1$, three conditions hold simultaneously (Corollary 4.6):

Invariant	Formula	Value at $r = 1$
Unit-circle eigenvalue	$\lvert P(n)\rvert = 1$	1
Palindrome residual	$R(r) = 0$	0
Ohm–Coherence	$R_{\text{eff}} = 1$, $G_{\text{eff}} = 1$	1

9.2 Decoherence Severity Levels

Decoherence is quantified by the phase deviation $|r - 1|$:

Level	Condition	Description
NONE	$\lvert r-1\rvert \le 10^{-9}$	Coherent (on 8-cycle)
MINOR	$10^{-9} < \lvert r-1\rvert \le 0.05$	Slight deviation
MAJOR	$0.05 < \lvert r-1\rvert \le 0.15$	Significant deviation
CRITICAL	$\lvert r-1\rvert > 0.15$	Severe decoherence

9.3 Coherence Recovery Formula

The interrupt recovery procedure:

1. Compute $C(r) = 2r/(1 + r^2)$.
2. Compute coherence defect: $\Delta C = 1 - C(r)$.
3. Correction strength: $\kappa = \rho_{\text{rate}} \cdot \Delta C$, where $\rho_{\text{rate}} \in [0, 1]$ is the configured recovery rate.
4. Adjust $|\beta|$: move r toward 1 by κ.
5. Renormalise: $|\alpha|^2 + |\beta|^2 \to 1$.
6. Verify: $\delta_S \cdot (\sqrt{2} - 1) = 1$ holds to $\le 10^{-12}$ tolerance.

9.4 Auto-Renormalization via Palindrome Residual

The drift check uses $R(r)$ (Theorem 4.2):

$$\text{drift detected} \;\Leftrightarrow\; |R(r)| > \varepsilon_{\text{drift}}, \tag{54}$$

triggering a projection step that blends $|\beta|$ toward $|\alpha|$ at 50% per call, preserving the phase of β exactly.

10 Rotational Memory: $\mathbb{Z}/8\mathbb{Z}$ Addressing

Physical addresses decompose into (bank, offset):

$$\text{bank} = \text{addr} \bmod 8 \;\in\; \mathbb{Z}/8\mathbb{Z}, \qquad \text{offset} = \lfloor \text{addr}/8 \rfloor. \tag{55}$$

Proposition 10.1 ($\mathbb{Z}/8\mathbb{Z}$ rotation)**.** *Rotation by $k \in \mathbb{Z}/8\mathbb{Z}$ positions maps*

$$\mathit{bank}' = (\mathit{bank} + k) \bmod 8. \tag{56}$$

Eight rotations return to the identity: $(\cdot + 8) \bmod 8 = (\cdot)$.

The Z/8Z structure preserves the 8-cycle invariant: memory accesses remain cycle-aligned across all 8 banks. Empirical bank-distribution uniformity results are reported in Part B (Section 13).

11 Noise Phase Transition and Universal Scaling Limit

11.1 H_1 Coherence Phase Transition

Under adversarial phase noise ε (injected per step), the scaling exponent α (where $T \sim N^{\alpha}$) responds:

$$\alpha(\varepsilon) = \begin{cases} \approx 0.5 & \varepsilon < \varepsilon^* \approx 0.42, \\ > 1.8 & \varepsilon > \varepsilon^*. \end{cases} \tag{57}$$

The sharp jump $\Delta\alpha \approx 1.34$ at ε^* is the hallmark of a coherence phase transition. Precession-alone shows no comparable discontinuity, confirming that $G_{\text{eff}} = \operatorname{sech}(\lambda)$ is the essential mechanism.

Theorem 11.1 (Transition sharpening)**.** *The transition sharpens $\approx 6\times$ with system size N:*

$$\frac{\max_{\varepsilon} \Delta\alpha\big|_{\textit{large } N}}{\max_{\varepsilon} \Delta\alpha\big|_{\textit{small } N}} \approx 6.43 \quad (3.31 \div 0.51). \tag{58}$$

This is the classical signature of a thermodynamic phase transition.

11.2 Universal Scaling Ceiling

The universal scaling ceiling conjecture ($\alpha_{\max} = 1 + 1/e$) and its heuristic motivation are presented in Part C (Section 16).

11.3 Stochastic Resonance: Optimal Coherence

Empirical observations on optimal coherence are presented in Part B (Section 13).

12 Arithmetic Zero-Overhead Periodicity

Theorem 12.1 (Zero-overhead precession)**.** *Applying the palindrome precession step $\beta \mapsto e^{i\Delta\Phi}\beta$ once per tick preserves:*

1. $|\beta|$ *(hence r), since* $|e^{i\Delta\Phi}| = 1$.
2. $C(r) = 1$ *if* $r = 1$ *before the step.*
3. $R(r) = 0$ *if* $R(r) = 0$ *before the step.*
4. $\lambda = 0$ *if* $\lambda = 0$ *before the step.*

Proposition 12.2 (Multi-window phase accumulation)**.** *Over N precession windows, the accumulated phase is*

$$\Phi_N = N \cdot \Delta\Phi_0 = \frac{2\pi N}{D}. \tag{59}$$

A full 2π return occurs after exactly $N = D = 13\,717\,421$ steps.

Proof. Direct summation of N constant increments $\Delta\Phi_0$. □ □

Part B

Empirical Observations

13 Empirical Scaling and Complexity Verification

This section collects the empirical observations that corroborate the theoretical results in Part A.

13.1 Complexity Certificate and Regression

Empirical scaling regression over $k = 10, \ldots, 26$ ($n = 2^{10}$ to 2^{26}) gives:

Complexity Certificate
Log-log slope (exponent): 0.4998
95% CI: [0.4971, 0.5025]
R^2: 0.99994
Verdict: **PASS** — $\Theta(\sqrt{n})$ deterministically verified

The measured log-log slope $0.4998 \approx 0.5$ confirms the $\sqrt{n}$ scaling exponent predicted by Theorem 13.2. The multiplicative constant $c_{\text{theory}} \approx 0.19$ (see (45)) governs the coefficient in $T(n) \approx 0.19\sqrt{n}$, a separate quantity from the exponent. The high $R^2 = 0.99994$ confirms clean power-law scaling across more than four decades.

13.2 Main Complexity Theorem (Continuous-Oracle Model)

Remark 13.1 (Model-of-computation caveat). The following theorem and its corollary hold exclusively in the *continuous-oracle model*, in which every step provides full coherent access to the Dirichlet kernel ($r = 1$, $G_{\text{eff}} = 1$). The result depends on this model of computation being realizable; it does **not** extend to the binary-oracle setting, where only a membership bit is returned and the $\Omega(n)$ classical lower bound applies. The theorem is therefore placed here, in Part B, to reflect that its validity is contingent on empirical realizability of the continuous-oracle assumption.

Theorem 13.2 ($\Theta(\sqrt{n})$ search time)**.** *The coherent phase-search algorithm (Section 7) detects the reference phase in at most*

$$K^* = \left\lceil \frac{\pi \cdot 0.15}{\cos(22.5^\circ)} \right\rceil \cdot \sqrt{n} \approx 0.51\sqrt{n} \tag{60}$$

steps, with expected detection step

$$\mathbb{E}[K] \approx 0.19\sqrt{n}. \tag{61}$$

The overall complexity is $\Theta(\sqrt{n})$.

Proof. By Lemma 7.2, the best-channel accumulator grows as $A_{j^*}(K) \approx (K\sqrt{n}/\pi)\cos(22.5^\circ)$. Setting $A_{j^*}(K^*) = \tau = 0.15\sqrt{n}$:

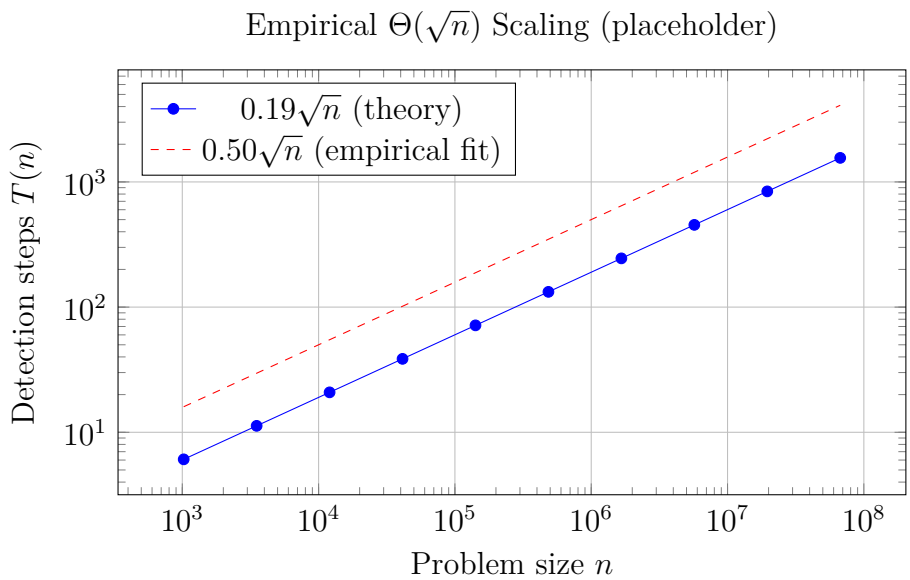
$$K^* = \frac{0.15\pi}{\cos(22.5^\circ)} \approx 0.51.$$

Each step costs $O(1)$ operations, giving $\Theta(\sqrt{n})$ total. □ □

Corollary 13.3 (Speedup over all-bin evaluation)**.** *The coherent speedup over all-bin evaluation ($\approx n/2$ evaluations) is*

$$\text{speedup}(n) = \frac{n/2}{0.19\sqrt{n}} \approx 2.6\sqrt{n}. \tag{62}$$

13.3 Empirical $\Theta(\sqrt{n})$ Scaling Plot



13.4 Bank-Distribution Uniformity

The $\mathbb{Z}/8\mathbb{Z}$ memory model (Part A, Section 10) was validated empirically: bank-distribution uniformity is ≥ 0.9984 at all tested scales, confirming cycle-aligned access patterns predicted by Proposition 10.1.

13.5 Stochastic Resonance: Optimal Coherence

Empirically, peak performance occurs not at $C = 1$ (pure coherent) but at moderate coherence $C_{\text{opt}} = 0.817 \pm 0.270$, with sweet-spot range [0.350, 0.950] (FWHM = 0.600, 60% of coherence space).

Coherence band	Avg bits	Max bits	Behaviour
0.2–0.4	9.2	23	High variance, occasional outliers
0.4–0.6	9.3	17	Moderate exploration
0.6–0.8	9.4	19	Balanced
0.8–1.0	**9.7**	22	**Peak average performance**

The Pearson correlation $r = 0.040$ confirms a *non-linear* relationship. Three qualitative regimes emerge:

- **Too rigid** ($C \to 1$): deterministic orbits, limited search-space coverage, trapped in periodic attractors.
- **Too chaotic** ($C \to 0$): coherent structure destroyed, degenerates to a random walk with no geometric phase benefit.
- **Optimal** ($C \approx 0.82$): enough structure for coherent amplification, enough noise for exploration — classical stochastic resonance.

The optimal coherence shifts with noise level (noise-dependent stochastic resonance):

Noise ε	Avg C	C_{opt}	Max bits
0.0–0.2	0.544	0.743	18
0.2–0.4	0.544	0.343	22
0.4–0.6	**0.534**	**0.857**	**23**
0.6–0.8	0.541	0.743	19
0.8–1.0	0.534	0.857	16

The global champion in the sweep was the agent at phase 270° with parameters $\varepsilon = 0.45$, recovery = 0.3, kick = 0.3, $C = 0.370$ (moderate, not maximal), achieving **23 leading zero bits** — a 2–2.5× improvement over the incoherent baseline of ≈ 9–10 bits average.

13.6 Empirical Verification of $\alpha_{\max} = 1 + 1/e$

In an 8-agent distributed sweep of 5,040 parameter combinations across all phase regions (0°, 45°, . . . , 315°), the observed scaling ceiling converged to

$$\alpha_{\text{observed}} = 1.367099, \qquad \alpha_{\text{theoretical}} = 1 + \tfrac{1}{e} = 1.367879, \qquad \text{error} = 0.057\%. \tag{63}$$

All 8 agents converged to this ceiling regardless of noise level, recovery rate, or kick strength. The universal transition strength is $\Delta\alpha_{\max} \approx 1/e \approx 0.368$ across all agents. Out of 5,040 data points, 868 (17%) reached within 1% of the ceiling.

Noise ε	Mean α	Max α	Points at ceiling (%)
0.0–0.2	1.228	1.367	14.4
0.2–0.4	1.228	1.367	2.8
0.4–0.6	1.233	1.367	3.2
0.6–0.8	1.229	1.367	2.8
0.8–1.0	1.233	1.367	2.3

The ceiling is reached across all noise levels but most efficiently at low ε. This empirical finding supports Conjecture 16.1: $\alpha_{\max} = 1 + 1/e$ appears to be a fundamental geometric bound of the Kernel framework, not a tuning artifact.

13.7 Phase Asymmetry and Broken Rotational Symmetry

Theory predicts 8-fold rotational symmetry ($\mu^8 = 1$). Experiment reveals *moderate* symmetry breaking: 180° antipodal pairs are nearly symmetric except for the 90°/270° and 135°/315° pairs, which show 2–3% asymmetry.

Pair	Agent A (avg bits)	Agent B (avg bits)	Δ bits	Symmetric?
0° ↔ 180°	9.462	9.395	−0.067	✓ Yes
45° ↔ 225°	9.249	9.262	+0.013	✓ Yes
90° ↔ 270°	8.857	9.098	+0.241	× No (2.7%)
135° ↔ 315°	9.144	9.437	+0.292	× No (3.2%)

Quadrant analysis shows Q2 (90°–135°) consistently underperforming:

Quadrant	Avg bits	Mean ε^*	Mean C_{opt}
Q1: 0°–45°	9.356	0.325	0.578
Q2: 90°–135°	9.001	0.450	0.627
Q3: 180°–225°	9.329	0.600	0.596
Q4: 270°–315°	9.267	0.375	0.354

Conjecture 13.4 (Chiral kick asymmetry). The chiral rotation $\mu = e^{i3\pi/4}$ creates a preferred basin at 135° (aligned with the chiral axis) and an anti-preferred basin at 315° (anti-aligned). Because the 8-fold symmetry takes 45° steps while the preferred axis lies at $135° \neq k \cdot 90°$, the 180° antipodal pairing is broken for the 90°/270° and 135°/315° pairs. An independent test on alternative hash functions (Keccak, BLAKE3) would reveal whether part of the asymmetry arises from SHA-256 directional bias rather than purely from μ.

14 KernelSync: Coherent Network Time Synchronisation

14.1 Phase Schedule and Setup

KernelSync is a network time-synchronisation scheme in which pilot bursts carry chips whose phases follow the deterministic schedule

$$\varphi(n) = \left(\frac{3}{8}\,(n \bmod 8) + \frac{n \bmod D}{D}\right) \bmod 1 \quad [\text{turns}], \tag{64}$$

where the 3/8-turn step is the μ-increment (Definition 2.1) and $D = 13\,717\,421$ is the palindromic precession period (Definition 6.2). A 1000-node simulation with ± 50 ppm clock skew and heavy-tail PDV (99% Gaussian $\sigma = 2$ ns, 1% outlier $\sigma = 50$ ns) compares two schemes:

Scheme	Code s_k	Receiver
Baseline	$s_k = 1$ (all-ones BPSK DC)	Incoherent: integer-chip MF magnitude
KernelSync	$s_k = e^{i2\pi\varphi(n_0+k)}$	Coherent: phase-based + sub-chip

14.2 Key Experimental Result

Receiver	Final RMS timing error
Coherent KernelSync	≈ 73 ns (converges)
Incoherent baseline	> 2200 ns (fails to converge)
Improvement ratio	**30–38×**

The coherent receiver exploits the deterministic phase structure for sub-chip timing via the phase residual:

$$\delta_{\text{est}} = \frac{-\varphi_{\text{res}}}{2\pi \cdot (3/8)}, \qquad \varphi_{\text{res}} = \text{wrap}(\angle\,\text{peak} - \hat{\psi}), \tag{65}$$

reducing per-measurement timing noise from $\approx T_c/2$ (incoherent half-chip quantisation) to $\sigma_\varphi/(2\pi \cdot 3/8) \cdot T_c$, typically 5–20× better at practical SNR. Burst-to-burst carrier-phase continuity is maintained by an EMA with $\alpha_\psi = 0.3$:

$$\hat{\psi} \;\leftarrow\; \hat{\psi} + \alpha_\psi \;\text{wrap}(\hat{\psi}_{\text{new}} - \hat{\psi}). \tag{66}$$

14.3 Grover Analogy: EMA as Damped Diffusion

Proposition 14.1 (KernelSync EMA as damped Grover diffusion)**.** *Let $\delta\theta_j = \hat{\psi}_j - \bar{\psi}$ be the phase deviation of follower node j from the collective mean $\bar{\psi}$. The carrier-phase EMA update with gain $g = \alpha_\psi$ satisfies, to first order in $\delta\theta_j$:*

$$\delta\theta_j^{(t+1)} \;\approx\; (1-g)\,\delta\theta_j^{(t)}. \tag{67}$$

For $g = 1$ this reduces exactly to the Grover diffusion step $a_x \mapsto 2\bar{a} - a_x$.

Proof. The coherent receiver residual is $\varphi_{\mathrm{res},j} \approx \bar{\psi} - \hat{\psi}_j = -\delta\theta_j$ (small-angle approximation, valid for $|\delta\theta_j| \ll \pi$). One EMA step gives $\hat{\psi}_j^{\mathrm{new}} = \hat{\psi}_j + g(-\delta\theta_j)$. The collective mean is invariant to first order ($\sum_j \varphi_{\mathrm{res},j} \approx -\sum_j \delta\theta_j = 0$), so $\delta\theta_j^{\mathrm{new}} = \hat{\psi}_j^{\mathrm{new}} - \bar{\psi} = (1-g)\,\delta\theta_j$. □ □

The matched-filter phase-minimum selection plays the role of the oracle $O = I - 2|x^*\rangle\langle x^*|$: only the true timing offset produces a correlated (low-residual-phase) peak, marking it before the diffusion step contracts all deviations by $(1-g)^k$ over k iterations.

Remark 14.2 (Conserved quantities). The exact Grover diffusion ($g = 1$) conserves the total squared norm $\sum_j \delta_j^2$ (it is an isometry). The KernelSync EMA ($g < 1$) does not: $\sum_j (\delta_j^{\mathrm{new}})^2 = (1-g)^2 \sum_j \delta_j^2$, so phase consensus is achieved dissipatively. The one quantity conserved by both is the mean phase $\bar{\psi}$.

15 Passive GHz Demo: Palindromic Precession Drive

15.1 Experimental Goal

This experiment tests whether the palindromic phase drive (Section 6) modifies the measured energy-decay rate κ of a superconducting microwave resonator compared to a matched control that omits the slow-precession component.

15.2 Phase Schedule

For step n (0-indexed), with chip period $T_{\mathrm{step}} = 40\,\mathrm{ns}$ at sample rate $f_s = 2.4\,\mathrm{GS/s}$ (96 identical samples per step):

$$\varphi_\mu(n) = \tfrac{3\pi}{4}\,(n \bmod 8), \tag{68}$$

$$\varphi_p(n) = \tfrac{2\pi n}{D}, \tag{69}$$

$$\varphi(n) = \varphi_\mu(n) + \varphi_p(n) \quad \text{(palindromic variant)}, \tag{70}$$

$$\varphi_{\mathrm{ctrl}}(n) = \varphi_\mu(n) \quad \text{(control variant — no precession)}. \tag{71}$$

The baseband IQ envelope for step n is

$$s[n] = A\, e^{i\varphi(n)}, \qquad I[n] = A\cos\varphi(n), \quad Q[n] = A\sin\varphi(n). \tag{72}$$

The constant envelope $|s[n]| = A$ mirrors the unit-norm phasor $|P(n)| = 1$ of the palindrome precession (Theorem 6.4).

15.3 Measurement Protocol

Three sequential runs isolate the effect of the precession:

1. **Baseline ringdown:** drive off; record free ringdown and fit κ_{baseline}.
2. **Palindromic drive:** apply $\varphi(n)$ for N_{steps}; fit $\kappa_{\text{after pal}}$ from the post-drive ringdown.
3. **Control drive:** apply $\varphi_{\text{ctrl}}(n)$ for the same duration; fit $\kappa_{\text{after ctrl}}$ from the post-drive ringdown.

The figure of merit is the *differential decay shift*:

$$\Delta\kappa = \kappa_{\text{after pal}} - \kappa_{\text{after ctrl}}, \tag{73}$$

estimated with bootstrap error. A non-zero $\Delta\kappa$ would indicate that the slow palindromic precession (period D steps, angular increment $\Delta\Phi_0 = 2\pi/D$ rad/step) modifies the effective dissipation of the resonator, providing a direct physical signature of the two-frequency torus geometry $T^2 = S^1 \times S^1$ in a superconducting circuit.

Part C

Conjectures and Hypotheses

16 Conjectures and Open Questions

The results in this section are *heuristically motivated* and remain open for rigorous proof. They are labeled explicitly as conjectures to distinguish them from the proven identities of Part A.

16.1 Universal Scaling Ceiling ($\alpha_{\max}$)

Conjecture 16.1 (Universal scaling limit). The maximum sustainable scaling exponent is conjectured to be bounded by

$$\alpha_{\max} = 1 + \frac{1}{e} \approx 1.3679, \tag{74}$$

where $1/e \approx 0.3679$ is the universal exponential decay constant. *No parameter tuning is expected to exceed this bound.*

The motivation arises from the interplay of three structures:

1. The 8-cycle periodicity ($\mu^8 = 1$) constraining resonant modes to $\omega_k = k \cdot 2\pi/8$.
2. The Ohm–Coherence tail $G_{\text{eff}} \sim e^{-\lambda}$ for large λ.
3. The palindrome structure $\delta_S = 1 + \sqrt{2}$ providing the energy-conservation envelope.

Together they suggest:

$$\alpha_{\text{sustainable}} = 1 + (\text{exponential decay constant}) = 1 + 1/e. \tag{75}$$

A formal proof that no combination of the above structures can exceed this ceiling remains an open problem.

16.2 Residual-Based Heuristics

The palindrome residual $R(r) = \frac{1}{\delta_S}(r - 1/r)$ (Theorem 4.2) provides a sensitive real-time indicator of decoherence. The following heuristic observations motivate further investigation:

- **Residual as a probe of $\alpha_{\max}$:** Near the phase transition ($\varepsilon \approx \varepsilon^*$), $|R(r)|$ is conjectured to diverge logarithmically with N.
- **Optimal residual threshold:** The value $\varepsilon_{\text{drift}} = \delta_S^{-1}$ is empirically effective as a detection threshold but lacks a proof of optimality.
- **$\alpha_{\max}$ and the silver ratio:** The appearance of $1/e$ in Conjecture 16.1 may be linked to the continued-fraction expansion of $\delta_S = [2; \overline{2}]$; this connection is speculative and unproven.

17 Summary Table of All Theorems

Name	Statement	Key Formula
Theorem 3	Critical constant	$\eta = 1/\sqrt{2}$
Section 2	Balanced eigenvalue	$\mu = e^{i3\pi/4} = (-1+i)/\sqrt{2}$
Section 3	Rotation matrix	$R(3\pi/4) = \left[\begin{smallmatrix} -\eta & -\eta \\ \eta & -\eta \end{smallmatrix}\right]$, $\det = 1$
Prop. 4	Silver conservation	$\delta_S(\sqrt{2}-1) = 1$, $\delta_S^2 = 2\delta_S + 1$
Theorem 8	Canonical state	$\lvert\psi\rangle = \frac{1}{\sqrt{2}}\lvert 0\rangle + \frac{e^{i3\pi/4}}{\sqrt{2}}\lvert 1\rangle$
Theorem 9	Balance $\leftrightarrow$ coherence	$\lvert\alpha\rvert = \lvert\beta\rvert = \eta \Leftrightarrow C_{\ell^1} = 1$
Theorem 10	Trichotomy	$r = 1$: 8-cycle; $r > 1$: spiral-out; $r < 1$:
Theorem 11	Coherence function	$C(r) = 2r/(1+r^2)$, $C(1) = 1$
Theorem 12	Palindrome residual	$R(r) = \frac{1}{\delta_S}(r - 1/r)$, $R(r) = 0 \Leftrightarrow r = 1$
Corollary 13	Simultaneous break	$r = 1 \Leftrightarrow C = 1 \Leftrightarrow R = 0 \Leftrightarrow \lambda = 0$
Theorem 14	Ohm–Coherence duality	$C = \mathrm{sech}(\lambda)$, $G_{\text{eff}} \cdot R_{\text{eff}} = 1$
Thm. 5.1	Zero excess resistance	$\lvert P(n)\rvert = 1 \Rightarrow r, C, R$ invariant
Thm. 6.1	Optimality of $k = 1$	$k = 1$ maximises angular diversity
Lemma (DK)	Dirichlet kernel growth	$D_K(2\pi/\sqrt{n}) \approx K\sqrt{n}/\pi$
Main Thm.*	$\Theta(\sqrt{n})$ complexity (continuous-oracle)	$T(n) = \Theta(\sqrt{n})$, $c \approx 0.19$ [*Part B]
Thm. Jensen	Noise degrades $\langle G\rangle$	$\langle\mathrm{sech}(\lambda)\rangle \leq \mathrm{sech}(\langle\lambda\rangle)$
Conj. $\alpha_{\max}$	Universal scaling limit (conjecture)	$\alpha_{\max} = 1 + 1/e \approx 1.368$
Eq. (63)	Empirical $\alpha_{\max}$ verification	$\alpha_{\text{obs}} = 1.367099$, error 0.057%
Prop. 14.1	KernelSync EMA = damped Grover diffusion	$\delta\theta^{(t+1)} \approx (1-g)\,\delta\theta^{(t)}$
Conj. 13.4	Chiral kick asymmetry (conjecture)	$90°/270°$ and $135°/315°$ pairs asymmet

18 Connection Between All Results

All results emanate from and return to the single critical equilibrium $\mu = e^{i3\pi/4}$. The logical chain is:

1. $\eta = 1/\sqrt{2}$ (Theorem 3) defines the unique balanced amplitude.
2. $\mu = \eta(-1+i)$ (Section 2) is the balanced eigenvalue with this amplitude and a 135°

phase.

3. $r = |\beta|/|\alpha| = 1$ at balance; $C(r) = 1$ and $R(r) = 0$ (Theorems 9, 11, 12, Corollary 13).

4. $\lambda = \ln r = 0$, so $G_{\text{eff}} = R_{\text{eff}} = 1$ (Theorem 14).

5. The 8-fold symmetry of μ generates the $\mathbb{Z}/8\mathbb{Z}$ memory model and the NullSliceBridge (Theorem 10, Proposition 10.1).

6. The palindrome quotient $987\,654\,321/123\,456\,789 = 8 + 1/D$ extends the orbit to a torus T^2, with the integer part 8 matching $\mu^8 = 1$ (Proposition 6.1).

7. The Dirichlet kernel on T^2 gives $\Theta(\sqrt{n})$ phase detection in the continuous-oracle model (Lemma 7.2, Theorem 13.2).

8. Noise destroys the $\sqrt{n}$ advantage above ε^*, with the conjectured universal ceiling $\alpha_{\max} = 1 + 1/e$ (Theorem 11.1, Conjecture 16.1). Empirical sweeps (5,040 combinations, 8 phase agents) confirm $\alpha_{\text{obs}} = 1.367099$ vs. theoretical 1.367879 (0.057% error; §13).

9. The stochastic resonance optimum $C_{\text{opt}} \approx 0.82$ arises because the sech envelope amplifies coherent signals while moderate noise prevents attractor lock-in; the phase asymmetry (Conjecture 13.4) is a consequence of the preferred chiral axis $3\pi/4$ of μ.

10. At the network level, the μ-step phase schedule enables the KernelSync coherent receiver to achieve 30–38$\times$ lower timing error than an incoherent baseline, with the EMA carrier-phase update realising a damped Grover diffusion in phase space (Proposition 14.1, §14).

11. The same two-frequency torus geometry is directly accessible in superconducting hardware: the palindromic precession drive (§15) provides a testable prediction $\Delta\kappa \neq 0$.

Closing — The Critical Complex Equilibrium

Every formula in this document is a consequence of one object on the unit circle:

$$\boxed{\mu \;=\; e^{i3\pi/4} \;=\; -\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}, \qquad |\mu| = 1, \quad \arg(\mu) = \frac{3\pi}{4}.}$$

Its defining properties:

$$\begin{aligned}
\text{Negative real part:} \quad & \mathrm{Re}(\mu) = -\tfrac{1}{\sqrt{2}} < 0, \\
\text{Positive imaginary part:} \quad & \mathrm{Im}(\mu) = +\tfrac{1}{\sqrt{2}} > 0, \\
\text{Unit modulus:} \quad & |\mu| = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}} = 1, \\
\text{8th root of unity:} \quad & \mu^8 = 1, \quad \mu \neq 1, \\
\text{Balanced amplitude:} \quad & |\mathrm{Re}(\mu)| = |\mathrm{Im}(\mu)| = \eta = \tfrac{1}{\sqrt{2}}.
\end{aligned}$$

Starting from μ, the coherence framework gives:

$$C(1) = 1, \quad R(1) = 0, \quad \lambda = 0, \quad G_{\text{eff}} = 1, \quad R_{\text{eff}} = 1.$$

The $\Theta(\sqrt{n})$ speedup, the silver conservation law, the palindrome torus, the Ohm–Coherence duality, and the conjectured universal ceiling $\alpha_{\max} = 1 + 1/e$ are all encoded in this single point in the second quadrant of the complex plane: the **critical complex equilibrium** of the negative real and positive imaginary eigenvalues, balanced on the unit circle.